MILITARY SPECIFICATION

TELEVISION DISPLAY REQUIREMENTS FOR THE WEAPON SYSTEM FOR GUIDED WEAPON MARK 1 MOD 0 AND MOD 2 (WALLEYE)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - This specification states the general performance requirements for all aircraft television displays used in conjunction with Guided Weapon Mk 1 Mod 0 and Mod 2 (WALLEYE). Only WALLEYE television display requirements are covered; no other requirements for alternate modes of presentation (radar displays and modified television displays) are specified.

1.2 Classification - The WALLEYE television displays shall be of the following types:

Type I - Modified radar displays
Type II - Modified television displays
Type III - WALLEYE display only

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military -

MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment

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MIL-M-7793	Meter, Time Totalizing
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Military -

MIL-STD-461	Electromagnetic Interference Characteristics - Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-756	Reliability of Weapons Systems, Procedures for Prediction and Reporting Prediction of
MIL-STD-781	Reliability Tests - Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-8	Versatile Avionics Shop Test System, Avionics Systems Compatibility, General Requirements for
AR-9	Versatile Avionics Shop Test System Test Program, General Requirements for
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for

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PUBLICATIONS

Military handbooks -

MIL-HDBK-217	Reliability Stress and Failure Rate Data for Electronic Equipment
MIL-HDBK-472	Maintainability Prediction

Naval Air Systems Command

EI-443	Avionics Installation Instructions for WALLEYE Guided Weapon System

2.2 Precedence of documents - When requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract: The contract shall have precedence over any specification.

(b) This specification: This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced specifications: Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

2.3 Availability of documents -

(a) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3. REQUIREMENTS

3.1 Preproduction testing - This specification makes provision for preproduction testing.

3.2 Parts and materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(a) Microelectronic items shall conform to requirements specified herein.

(b) Other parts and materials requirements shall conform to MIL-E-5400.

(c) Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Non-repairable subassemblies must be reliable. (See 6.4.)

(d) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard parts and materials approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

3.3 Design and construction - The equipment shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total weight - Total weight of the equipment, excluding cables, shall be a minimum consistent with good design.

3.3.2 Reliability - **The** Mean- (operating) Time-Between-Failure (MTBF) requirement for the television display shall be met when tested in accordance with 4.4.3 of this specification. All television display design activities shall be aimed at an objective of achieving an inherent reliability of the television display equal to the nominal (Design Goal) MTBF requirement. Reliability predictions, in full accordance with MIL-STD-756 and the "detailed stress analysis prediction procedure" of MIL-HDBK-217, shall be utilized as design tools in achievement of the nominal (Design Goal) MTBF requirement. Every economically feasible effort shall be exercised during production to cause a reliability growth toward the nominal MTBF requirement.

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3.3.2.1 **Minimum MTBF - The Mean (Operating) Time Between Failures (MTBF) shall be at least 135 operating hours when demonstrated in accordance with 4.4.3 herein.**

3.3.2.2 Nominal MTBF (design goal) - The nominal MTBF shall be 935 operating hours in an airborne environment.

3.3.3 Operational stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of at least 135 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.4 Operating life - The equipment shall have total operating life of 2,500 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.5 Time totalizing meter - The equipment shall contain a time totalizing meter in accordance with MIL-M-7793.

3.3.6 Cabling and connections -

3.3.6.1 Cables and connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400 and EI-443.

3.3.6.2 Interconnection cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of EI-443.

3.3.7 Interchangeability - Initial designs for the Type I and Type II displays shall be physically interchangeable with unmodified indicators of the same type. Initial designs for the Type III displays shall meet the interchangeability and replaceability requirements of MIL-I-8500.

3.3.7.1 Interchangeability of reordered equipment - For reordered equipment, interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously *manufactured equipment, supplied or designated by the procuring activity.*

3.3.8 Interference control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-STD-461.

3.3.9 **Provisions for Maintainability - The Maintainability Program built-in test features~~, construction~~ and packaging, provisions for test points, and other maintainability parameters shall be as specified in Specification AR-10.**

3.3.9.1 Mean-time-to-repair (MTTR) - The Mean-Time-To-Repair (MTTR) of the equipment shall be no greater than 2 hours when faulty modules,

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assemblies, and parts intended for replacement are replaced at the intermediate level. A prediction of MTTR shall be made in accordance with MIL-HDBK-472 Procedure III.

3.3.9.2 Compatibility with VAST - The equipment shall be compatible with the Versatile Avionic Shop Test System (VAST) and shall meet the requirements of Specification AR-8. When required by contract VAST Test Programs shall be furnished in accordance with Specification AR-9. If VAST Test Programs exist for the equipment, and changes to the equipment are made which will affect the fault diagnosis procedure, changes to the existing Test Program shall be prepared as part of the equipment changes in accordance with Specification AR-9.

3.3.10 Nomenclature and nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.11 Service conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 aircraft equipment.

3.3.11.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of curve 1 of Specification MIL-E-5400.

3.3.12 Standard conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

(a) Temperature: Room ambient (25 $\pm$ 5° Centigrade (C)).

(b) Altitude: Normal ground.

(c) Vibration: None.

(d) Humidity: Room ambient up to 90 percent relative humidity.

(e) Input power voltage: 115.0 $\pm$ 1.0 volts, 400 $\pm$ 40 cps, and 27.5 $\pm$ 0.5 volts d-c.

3.3.13 Warm-up time- The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed 5 minutes under standard or extreme service conditions.

3.3.14 Primary input power requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when supplied with power with characteristics as defined in MIL-STD-704, Category B.

3.3.14.1 Degraded performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.15 Standby provisions - The equipment shall provide for a standby position in which rated voltage shall be applied to all tube filaments with no voltages applied to other parts of the equipment.

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3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under standard conditions will be specified. The parameters specified in the following paragraphs represent the minimum acceptable criteria for weapon system operation.

3.4.1 Operation - The television display will receive a composite video signal from the WALLEYE weapon and shall provide a raster and presentation which will enable the operator to identify the target and confirm weapon lock on. The WALLEYE composite video signal parameters are illustrated in figure 1.

3.4.2 Detail requirements - The television display shall meet the detail requirements specified herein.

3.4.2.1 Video characteristics -

3.4.2.1.1 Aspect ratio - The television display shall accept a composite video signal with a 1 to 1 aspect ratio having a ±10 percent tolerance on each dimension. The presentation shall have a horizontal dimension of not less than 3.5 inches.

3.4.2.1.2 Scan standards - The television display shall have a raster scan of 525 lines per frame, 2 fields per frame (interlaced 2 to 1), and 30 frames per second. Spacing between consecutive interlaced lines shall not vary more than 15 percent from the average line spacing.

3.4.2.1.3 Video amplifier bandwidth - The video amplifier bandwidth shall be flat within ±1.0 decibel (db) from 32 cycles per second (cps) to 6 megacycles per second (mc) and shall be down no more than 3 db at 16 cps and 8 mc. There shall be no peaks and valleys in the bandwidth greater than ±1.0 db. With a step function input the overshoot shall not be greater than 20 percent of the amplitude of the output step function voltage. These parameters shall be determined under the following conditions:

(a) The video amplifier shall be loaded with its normal operating capacitance load.

(b) The video level shall be 50 percent of the level required to give a 30 to 1 contrast ratio at 300 foot-lamberts maximum brightness when viewed through any light-absorbing filters which are a part of the television display.

3.4.2.1.4 Shades of gray, brightness, and contrast - The television display presentation shall be measured using an input signal with 13 logarithmic video steps, representing a gamma of 1.8 and a 100-to-1 brightness ratio. The display shall provide 12 visible steps with a reference white level of

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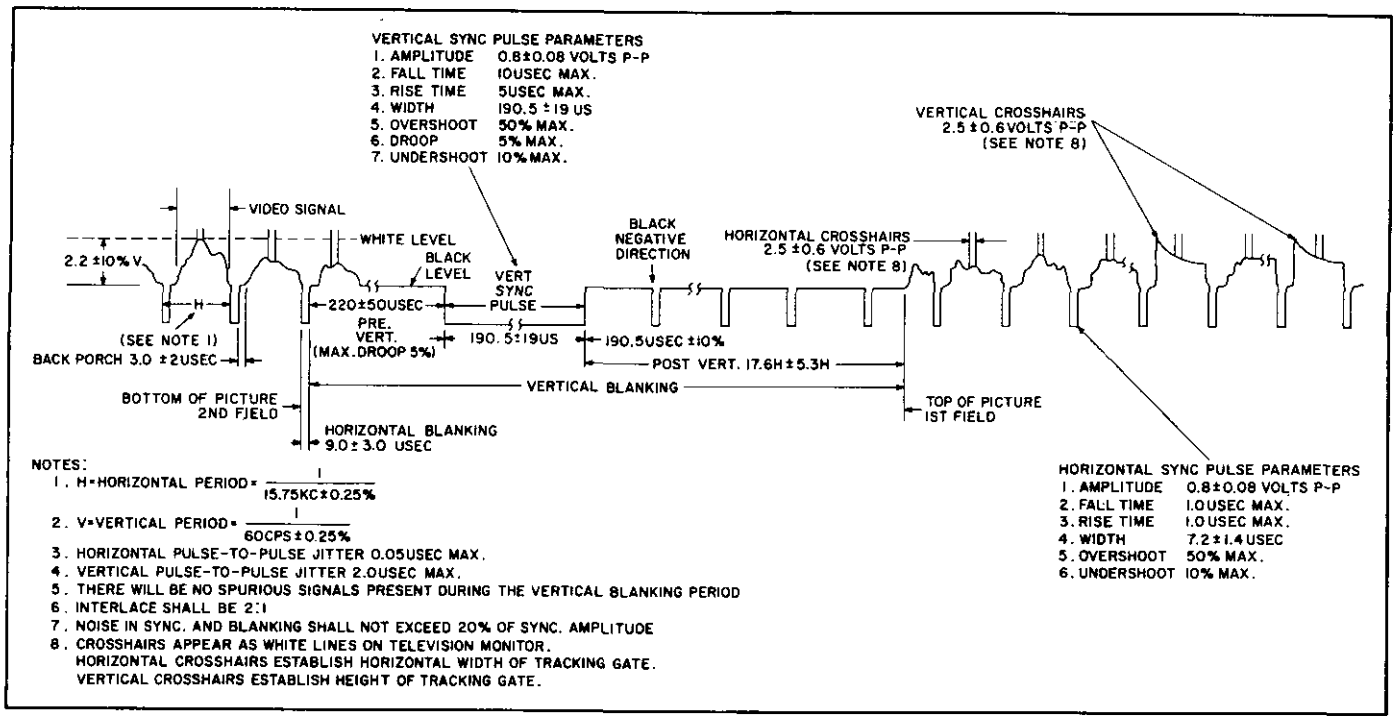


Figure 1. Composite video parameters

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300 foot-lamberts minimum without blooming when viewed through any light-absorbing filters which are a part of the television display. There shall be controls on the display to permit the operator to adjust both the brightness and contrast of the display.

3.4.2.1.5 Light interference - When flooded with uniformly diffused light of 3,000 foot-lamberts at an angle of incidence of 30 to 45 degrees, the television display shall have a brightness ratio of 15 to 1, minimum.

3.4.2.1.6 Display tube resolution - With a 450-line raster (approximately 9-mil spot size for a 4-by-4 inch display) the modulation of the horizontal raster lines shall be greater than 2 percent of the maximum black-to-white ratio in the center third of the tube. With the raster rotated 90 degrees relative to the tube face, the minimum response for the vertical raster lines shall be greater than 2 percent of the maximum black-to-white ratio.

3.4.2.1.7 No video input - With no composite video input, the television display shall have either a full visible raster or no raster.

3.4.2.1.8 Signal storage - Signal storage shall not exceed 20 percent from frame to frame when operated in the television mode.

3.4.2.1.9 Maximum white level video - The television display circuitry shall not be damaged by a maximum white-video level of 6.0 volts peak amplitude with a duration of 4.0 microseconds.

3.4.2.1.10 Edge effect - The television display circuitry shall not be damaged by an edge effect in the composite video which may result in peak white video of 8.0 volts amplitude. The duration of this edge effect shall be 0.5 microsecond maximum at the start and end of each horizontal line, and two lines at the start and end of each frame.

3.4.2.2 Vertical characteristics -

3.4.2.2.1 Frame rate - The television display shall have a frame rate of 30 frames (60 fields) per second.

3.4.2.2.2 Vertical retrace time - The television display shall have a vertical retrace time of not greater than 1,000 microseconds.

3.4.2.2.3 Vertical linearity - The vertical linearity deviation of the television display shall not exceed 5 percent.

3.4.2.2.4 Minimum vertical sync signal - With a vertical sync input of 0.6 volt peak-to-peak, the television display shall not exhibit vertical roll or interlace variation greater than that specified in 3.4.2.1.2.

3.4.2.3 Horizontal characteristics -

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3.4.2.3.1 Horizontal frequency - The television display shall be capable of satisfactory operation with a horizontal input frequency of 15,750 ± 50 cps.

3.4.2.3.2 Horizontal retrace time - The television display shall have a horizontal retrace time of not greater than 10 microseconds.

3.4.2.3.3 Horizontal linearity - The horizontal linearity deviation of the television display shall not exceed 5 percent.

3.4.2.3.4 Minimum horizontal sync signal - With a horizontal sync input of 0.6 volt peak-to-peak the television display shall not exhibit horizontal tearing. Phase jitter of the horizontal sweep shall not exceed 0.05 microsecond.

3.4.2.4 Input impedance - The input impedance of the television display shall be 93 ohms ± 10 percent. The video input to the television display shall float above system ground. Video signal ground for the weapon system shall be established in the WALLEYE weapon.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(a) Preproduction (first article) tests: (see 4.2).

(b) Initial production tests: (see 4.3).

(c) Acceptance tests: (see 4.4).

(d) Life test: (see 4.5)

4.2 Preproduction (first article) tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness

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or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (first article) test data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review.

4.2.2 Scope of tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract. Preproduction tests shall include environmental testing in accordance with MIL-T-5422, with modification of the test limits of the applicable tests to correspond with the limits specified herein. Interference tests shall be conducted in accordance with MIL-STD-461. Maintainability assurance tests shall be in accordance with AR-10.

4.2.3 Preproduction (first article) approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Manufacturing of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection. For equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial production tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on the equipment prior to starting the Initial Production Tests. The preproduction sample shall not be used for this test.

4.3.1 Scope of tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction

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sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Accessory material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial production sample approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of initial production test sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests
(b) Sampling Tests
(c) Reliability Assurance Tests
(d) Special Tests

4.4.1 Individual tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product
(b) Operational Test
(c) Manufacturing Run-In Test

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4.4.1.1 Examination of product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing run-in test - Each equipment shall be operated under the conditions specified herein for a period of at least 6 hours without failure. The minimum total manufacturing run-in test time shall be the burn-in period under conditions established by the Reliability Index Determination Tests or the Reliability Assurance Tests. Failures shall be reported and analyzed. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment will be permitted. The following conditions shall be used for the manufacturing run-in test.

(a) Temperature: Room ambient.

(b) Humidity: Room ambient.

(c) Vibration: Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of $\pm$3g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. During the period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The period of operation specified above may be divided into two periods to conform with standard working hours. This test is not required if a Reliability Assurance test is selected which includes a test on each equipment which consumes at least 10 hours of operation.

4.4.2 Sampling tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected

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for sampling tests by the Government inspector in accordance with table I. Sampling tests are not required when Reliability Assurance Tests are conducted.

Table I. Production sampling

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(a) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(b) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual, and special tests.

(c) Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment. This extended manufacturing run-in test shall be conducted only if the reliability assurance tests of 4.4.3 are deleted by the contract.

4.4.3 Reliability assurance tests - Reliability assurance tests shall be conducted as specified in MIL-STD-781 for both the Qualification and Production **Acceptance** Phases. Equipments selected for reliability assurance testing shall first have passed the individual and sampling tests.

4.4.3.1 Qualification phase - Prior to acceptance of equipments under the contract or order, a minimum of 3 equipments shall be tested as specified in MIL-STD-781 for the Qualification Phase. The maximum number of equipments to be tested shall be as specified in table 5 of MIL-STD-781. The Accept-Reject criteria for Test Plan III shall be used.

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4.4.3.2 Production acceptance phase - Each equipment shall be tested for 30 hours. Prior to the 30-hour test on each equipment a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof shall be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract, the operating test hours, and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled, and the results compared with the reject line of Test Plan III of MIL-STD-781. (Extend the line as necessary to accommodate the data.)

These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Department of the Navy, Washington, D. C. 20360 (Attention: Avionics Division).

At any time that the current totals of test hours and test failures plotted on Test Plan III curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists, pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 Test level - For both the Qualification and Production Acceptance Phases, Test Level F of MIL-STD-781, as modified herein, shall be used.

4.4.3.3.1 Temperature cycling - The heating portion of the temperature cycle shall be the time required for the equipment to stabilize at the high temperature limit plus 4 hours of operation as specified in 4.4.3.4 with the equipment stabilized at the high temperature limit. The cooling portion shall be as specified in MIL-STD-781.

4.4.3.4 Duty cycle - The equipment duty cycle shall consist of the following two modes:

Mode I: The equipment energized, with no signal inputs, from the beginning of the heating period to the beginning of Mode II;

Mode II: The equipment operated, following a typical mission operating profile, for a 4-hour period from the time of stabilization at the high temperature limit.

Accelerated reliability testing by a shortened duty cycle with operation increased to full stress is encouraged and recommended, provided the test plan is approved by the procuring activity.

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4.4.3.5 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.3.6 Equipment performance - The performance requirements specified in 3.4 shall be measured. The performance requirements shall be met at all temperatures from -54 to +71° Centigrade.

4.4.3.7 Failure criteria - In addition to the requirements of MIL-STD-781, equipment failure shall have occurred when performance does not meet the requirements of 3.4 with the inputs varied between specified extremes.

4.4.4 Special tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special test schedule - Selection of equipments for special tests shall be made as follows:

(a) On an early equipment after an engineering or material change.

(b) Whenever failure reports or other information indicate additional tests are required, as determined by the procuring activity.

4.4.4.2 Scope of tests - Special tests shall consist of those as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment failure - Should a failure occur during either the sampling, reliability assurance, or special tests, the contractor shall:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(d) Where practical, include a test in the individual tests to check all equipment for this requirement

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until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test shall be performed under the conditions specified in 4.5.1. Equipments which have successfully passed the initial production tests, sampling tests, reliability tests, or special tests may be selected for life tests. When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed in table II receive at least 300 hours each of test time. The life test sample shall be selected by the Government inspector in accordance with table II.

Table II. Life test sampling

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test conditions - The life test shall be conducted under the following simulated service conditions:

(a) Temperature: normal room.

(b) Altitude: normal ground (0 - 5000 ft.).

(c) Humidity: room ambient.

(d) A-c voltage: 115 ± 5 volts (at lowest applicable frequency).

(e) D-c voltage: 27.5 ± 2.0 volts.

4.5.2 Test periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

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4.5.4 Test data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(a) Part type number.

(b) Circuit reference symbol number.

(c) Part function.

(d) Name of the manufacturer.

(e) Nature of the failure.

(f) Number of hours which the part operated prior to failure.

4.5.4.1 Failure report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity on completion of the test, in which the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test procedures - Procedures used for conducting preproduction tests, acceptance tests, and life test shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of tested equipment - Equipment which has been subjected to initial production, acceptance, and life tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission testing - No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended use - The WALLEYE television display shall be suitable for installation and use in high-performance carrier-based or land-based aircraft. The television display is to be used in conjunction with Guided Weapon Mk 1 Mod 0 and Mod 2 (WALLEYE).

6.2 Test values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 $\pm$ 0.5 volts d-c and 115.0 $\pm$ 1.0 volts 400 $\pm$ 40 cps a-c as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standards of performance.

6.3 Performance objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance, or reliability will result from the use of materials, parts, and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 Nonrepairable subassemblies - As a general rule nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts, tolerances, and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high-voltage power supplies), the nonrepairable subassembly should evidence a mean-time-to-failure greater than 5,000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

(c) Data requirements.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-81624(AS) TELEVISION DISPLAY REQUIREMENTS FOR THE WEAPON SYSTEM FOR GUIDED WEAPON MK 1 MOD 0 ND MOD 2 (WALLEYE)

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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